

PRAGATI TOPPO

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EDUCATION

California Polytechnic State University, *Software Engineering*

06/2026

Relevant Coursework: Data Structures, Project-Based Object Oriented Programming and Design, Intro to Computer Organization, Discrete Structures, Individual Software Design and Development, Design and Analysis of Algorithms, Intro to Computer Graphics, Intro to Data Science, Software Engineering 1 & 2, Statistical Machine Learning (Indian Institute of Technology, Bombay), Cloud Computing with AWS (Indian Institute of Technology, Kanpur)

San Luis Obispo, United States

EXPERIENCE

Undergraduate Researcher (SURP), *Cal Poly, College of Engineering*

06/2025 – Present

- Developed uncertainty quantification methods for machine learning models analyzing drone-based coastal imagery to monitor eelgrass ecosystem dynamics in Morro Bay
- Enhanced existing binary classification framework by implementing probabilistic algorithms that convert internal model outputs into confidence intervals for spatial coverage measurements
- Conducted model validation and calibration through empirical comparison of predicted uncertainty against ground truth annotations to improve measurement credibility
- Contributed to advancing remote sensing techniques for quantifying temporal changes in critical marine habitat distributions using computer vision and statistical analysis

Software Engineer Intern, *Hexaware Technologies*

07/2024 – 09/2024

- Enhanced system security by optimizing database queries through indexing and query optimization, reducing transaction processing time by 68%
- Implemented performance testing protocols to identify and resolve system vulnerabilities and bottlenecks, improving overall system throughput by 25%
- Refactored REST API architecture using secure design principles, resulting in improved response times and reduced server resource utilization
- Monitored application performance and identified potential security weaknesses in system architecture

Instructional Student Assistant, *Cal Poly Computer Science and Software Engineering Department*

09/2023 – 06/2025

- Developed and implemented optimized grading rubrics that increased assessment efficiency by 30% while maintaining comprehensive feedback quality
- Created streamlined review materials for Data Structures (CSC 202) that improved student algorithm implementation performance by an average of 25%

SKILLS

Programming Languages

Python, C++, JavaScript, Java, C, TypeScript, HTML/CSS, SQL, R, MATLAB, CUDA, Bash/Shell

Software Tools and Technologies

Git, GitHub, Docker, Kubernetes, AWS, Linux, PostgreSQL, MySQL, MongoDB, OpenGL, GLSL, JPA, Postman, Figma

Framework and Libraries

React.js, Node.js, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Flask, Django REST Framework

PROJECTS

Spinterest, *CSC 308, Software Engineering I* 

10/2024 – 03/2025

- Designed Figma wireframe for a Pinterest-Spotify hybrid web application with customizable dashboard interface for organizing and converting music playlists
- Developed responsive profile page using JavaScript, Node.js and Material-UI to display user information and pinned music
- Integrated with Spotify API to enable seamless playlist management through RESTful API calls
- Created dynamic search controller with real-time database querying based on partial username inputs
- Implemented cross-profile content pinning feature and wrote Cypress test cases to ensure reliable functionality

- Successfully completed full database integration enabling all planned features

Disco Dance Floor Explorer, *CSC 471, Intro to Computer Graphics*

04/2025 – 06/2025

- Built 3D interactive application with animated dancing robots and narrative gameplay
- Implemented hierarchical animation system with 20+ joints for detailed character movement
- Developed Blinn-Phong lighting pipeline across multiple shader programs
- Created multi-modal camera system with free exploration and first-person perspective

File System Emulator, *CSC 357 - Systems Programming*

02/2024

- Developed a custom file system emulator in C using advanced data structures and memory management techniques
- Implemented core file system operations including directory navigation, content listing, directory creation, and regular file creation
- Engineered command-line interface that parses arguments to specify directory names, verifies existence, and loads inode list files
- Optimized memory usage through efficient struct design and proper allocation/deallocation to prevent memory leaks
- Applied low-level system programming concepts to simulate actual file system behavior and operations

Secure Web Server Implementation, *In Association with IIT Kanpur*

06/2024

- Designed and deployed two distinct subnets: a DMZ Network (publicly accessible) and a Private Network, ensuring robust security and segregation of public-facing and internal resources.
- Successfully launched and configured a Web Server in the DMZ subnet, making it securely accessible to the public.
- Implemented an RDS Server within the Private subnet, enhancing data security by restricting direct public access and configuring NAT for controlled access.

User Login API with Email and OTP Authentication

07/2024

Developed using Django REST Framework, PostgreSQL, and Docker

- Developed a secure backend API for user authentication using Django, featuring email-based user registration, OTP generation and verification for login, and session management with secure tokens.
- Implemented rate limiting, encrypted communication, and secure algorithms to enhance security.

LEADERSHIP EXPERIENCE

Tech Officer, *Women in Software and Hardware*

05/2025 – Present

- Plan technical workshops (soldering, 3D printing, web development) and career development events for members
- Collaborate with specialized clubs to expand learning opportunities in hardware, robotics, and app development
- Mentor underclassmen through academic and professional development since sophomore year

Team Tech Lead, *Society of Women Engineers*

11/2023 – 10/2024

- Led a team of 5 students in designing and developing a marketing website using PHP and CSS, focusing on front-end development to effectively showcase our Wheelchair Securement System for the Team Tech Competition sponsored by Boeing at WE24.
- Conducted research and extensive testing to create a Securement System for airplanes, aimed at reducing damage and misplacement of wheelchairs, improving accessibility for wheelchair users in air travel.